Lösung: Unbestimmte Integrale

a) $f(x) = x^n$ mit $n \neq -1$

$F(x) = \frac{1}{n+1} x^{n+1} + C$

..., da $F'(x) = \frac{n+1}{n+1} x^{n+1-1} + 0 = x^n = f(x)$ ✓

↑ keine Division durch 0, da $n \neq -1$

b) $f(x) = 3x^5 - 2x^3 + x^2 + 1$

$F(x) = \frac{3}{6}x^6 - \frac{1}{2}x^4 + \frac{1}{3}x^3 + x + C$

..., da $F'(x) = \frac{3}{6} \cdot 6x^5 - \frac{1}{2} 4 \cdot x^3 + \frac{3}{3} x^2 + 1 + 0$

$= 3x^5 - 2x^3 + x^2 + 1 = f(x)$ ✓

c) $f(x) = \ln|x|$

$F(x) = \ln|x| \cdot x - x + C$

..., da $F'(x) = (\ln|x| \cdot x)' - (x') + 0$

$= (\ln|x|)' \cdot x + \ln|x| \cdot (x)' - (x')$

$= \frac{1}{x} \cdot x + \ln|x| - 1 = 1 + \ln|x| - 1 = \ln|x| = f(x)$ ✓

d) $f(x) = 1$

$F(x) = x + C$

..., da $F'(x) = 1 + 0 = 1 = f(x)$ ✓

e) $f(x) = \frac{1}{\sqrt{x}} = x^{-1/2}$

$F(x) = \frac{1}{(+1/2)} x^{1/2} + C = 2\sqrt{x} + C$

..., da $F'(x) = 2 \cdot \frac{1}{2\sqrt{x}} = \frac{1}{\sqrt{x}} = f(x)$ ✓

e) $f(x) = e^{ax}$

$F(x) = \frac{1}{a} \cdot e^{ax} + C$

..., da $F'(x) = \frac{1}{a} \cdot (e^{ax})' + 0 = \frac{1}{a} \cdot a \cdot e^{ax} = e^{ax} = f(x)$ ✓

g) $f(x) = \cos(x) + \sin(x)$

$F(x) = \sin(x) - \cos(x) + C$

..., da $F'(x) = (\sin(x))' - (\cos(x))' + 0$

$= \cos(x) - (-\sin(x)) = \cos(x) + \sin(x) = f(x)$ ✓

h) $f(x) = \frac{5}{1+x^2}$

$F(x) = 5 \cdot \arctan(x) + C$

..., da $F'(x) = 5 \cdot (\arctan(x))' + 0 = 5 \cdot \frac{1}{1+x^2} = f(x)$ ✓

i) $f(x) = 2 \cdot \sqrt{x} = 2\, x^{1/2}$

$F(x) = \frac{2}{3/2}\, x^{3/2} + C = \frac{4}{3} x^{3/2} + C$

..., da $F'(x) = \frac{4}{3} \cdot \frac{3}{2} \cdot x^{1/2} + 0 = 2 \cdot x^{1/2} = 2\sqrt{x} = f(x)$ ✓

j) $f(x) = u'(x) \cdot v(x) + u(x)\, v'(x) = (u(x) \cdot v(x))'$

$F(x) = u(x) \cdot v(x) + C$

..., da $F'(x) = (u(x) \cdot v(x))' + (C)'$

$= u'(x) \cdot v(x) + u(x)\, v'(x) + 0 = f(x)$ ✓

k) $f(x) = \underset{u'(x)}{\cos(x)} \cdot \underset{v(x)}{x} - \underset{u(x)}{\sin(x)} \cdot \underset{v'(x)}{1} = (\sin(x) \cdot x)'$

$F(x) = \sin(x) \cdot x + C$

..., da $F'(x) = (\sin(x) \cdot x)' + (C)' = (\sin(x))' \cdot x + \sin(x) \cdot (x)' + 0$

$= \cos(x) \cdot x + \sin(x) \cdot 1 = f(x)$ ✓